BRAINARD PHILEMON JAGATI

+91 7993916209 | brainardjagati27@gmail.com | LinkedIn | GitHub | Google Scholar | Portfolio

EDUCATION

- **Indian Institute of Information Technology Nagpur** *2023 - 2027*
  Bachelor of Technology, Computer Science Engineering (Data Science and Analytics) Nagpur, India
  - CGPA: 8.85/10.00

EXPERIENCE

- **Nvtha AI** *Aug 2026 - Present*
  Lead Researcher (Hybrid) Singapore
  - Developed and evaluated hallucination detection methods for language models, using representation-based signals to identify unsupported outputs and adapt the approach for deployment.
  - Ran experiments to compare detection approaches, analyze failure cases, and improve robustness for real-world LLM workflows.

- **Singapore University of Technology & Design [🌐]** *July 2026 - Present*
  Visiting Student (On-site) Singapore
  - Conducting research with Prof. Ezekiel Soremekun on the Jailbreak Attacks & Defenses for Code Large Language Models.
  - Researching on white-box defense mechanisms for jailbreak attacks in open-source Code LLMs by exploiting embedding-level representations and internal model activations.

- **Nanyang Technological University [🌐]** *May 2026 - Present*
  Research Intern (Remote) Singapore
  - Conducting research with Prof. Erik Cambria on the safety of Multimodal Large Language Models (MLLMs) for medical applications, focusing on the interaction of clinical images and textual information.
  - Investigating hallucination mitigation mechanisms in medical MLLMs using RAG-based methods.

PUBLICATIONS

1. Brainard Philemon Jagati *et al.* (Sept 2026). **Semantic-Aware Logit Adjustment for Long-Tailed Recognition**. Manuscript accepted at *NewinML workshop @NeurIPS 2026*.
2. Brainard Philemon Jagati *et al.* (Feb 2026). **A Comprehensive Survey of Logit Adjustment Methods for Long-Tailed Recognition**. *Preprint*.
3. Brainard Philemon Jagati *et al.* (Jan 2026). **Class–Confidence Aware Reweighting for Long-Tailed Learning**. *Preprint*.

PROJECTS

- **Detecting Textual Backdoors through Internal Latent-Space Perturbations** [GitHub]
  Tools: Python, PyTorch, Transformers, NLP
  - Extended the ONION defense framework with Leave-One-Out (LOO) embedding shift and early-layer analysis to detect and mitigate textual backdoor triggers in large language models.
  - Built a poisoning-and-defense evaluation pipeline using clean and BadNets-style poisoned data (5% poisoning rate), with scripts for training a poisoned victim model and testing the defense on SST-2.
  - Reduced attack success rate from 16.34% to 11.95% while maintaining comparable clean accuracy (89.91% to 89.84%), with the defense fully self-contained inside the victim BERT model.

SKILLS

- **Programming Languages:** C, C++ (Proficient), Python, Java
- **Libraries/Frameworks:** TensorFlow, PyTorch, Scikit-learn, OpenCV
- **Tools:** MySQL, PowerBI, Tableau, Weka, Git, VSCode, Hugging Face
- **Relevant Coursework:** Probability & Statistics, Linear Algebra, Discrete Mathematics, Artificial Intelligence, Machine Learning, Computer Vision & Deep Learning
- **Quantitative Research:** Alpha Research & Signal Development, Factor Modeling, Cross-Sectional & Time-Series Analysis, Statistical Arbitrage, Backtesting
- **Research Interests:** Neural Collapse Theory, Long-Tailed Learning, Representation Learning, LLM Safety, Trustworthiness & Privacy.

ACHIEVEMENTS

- **NeurIPS 2026 Financial Assistance Award**
  Awarded to present accepted paper at the NewInML Workshop @ NeurIPS 2026, Paris
- **Expert - Codeforces** [Profile]
  (Peak Rating - 1612)
- **4 Stars - Codechef** [Profile]
  (Peak Rating - 1947)
- **Global Rank 70 in Codechef Starters 211 among 30,000+ participants.**
  Codechef
- **Global Rank 552 in Codeforces Round 1076 among 43,000+ participants.**
  Codeforces
- **Rank 46 out of 500+ participants in TrexSim Alpha Challenge**
  TrexQuant

POSITION OF RESPONSIBILITY

- **Founder**
  Code O' Phile
  ◦ Hosted nationwide coding quizzes and webinars attracting over 3000 students from various colleges.
- **Core Member**
  Dotslash (The Coding Club) - IIIT Nagpur
  ◦ Facilitated weekly competitive programming sessions and post contest discussions for juniors.